

PRODUCT REQUIREMENTS DOCUMENT

GameNest Hub

Online Gaming Portal

Web Development · Vanilla JS · Static Site · \$50–\$100 USD

This document defines the complete requirements for building GameNest Hub - a browser-based casual gaming portal delivering nine free HTML5 games across Puzzle, Arcade, and Trivia categories. All pages must open directly from index.html with zero server dependencies.

SECTION 01

Project Overview

GameNest Hub is a fully static browser gaming portal where visitors discover, play, and rate lightweight HTML5 mini-games no account needed. The platform targets students and office workers wanting 5–10 minute sessions. A localStorage-based leaderboard tracks session scores per game. The build uses only HTML5, CSS3, and vanilla JavaScript with no paid frameworks, no paid APIs, and no server-side code.



SECTION 02

Goals & Success Criteria

- Deliver a zero-backend portal openable from index.html.
 - Real-time catalog filter by category and keyword search.
 - localStorage leaderboard with ranked score display.
- Responsive at 375px, 768px, and 1280px breakpoints.
 - Dark gaming aesthetic using defined design tokens only.
 - All 5 pages load with zero console errors.

Out of Scope

- User authentication or cloud score storage.
 - Server-side rendering (Node, PHP, etc.).
- Paid APIs, analytics, or ad integrations.
 - PWA service workers or native mobile builds.

SECTION 03

Scope of Work

Page	File	Primary Responsibility
Homepage	index.html	Hero, 6 featured game cards, footer footer
Game Catalog	catalog.html	All 9 games, category filters, real-time keyword search
Game Detail	game-detail.html	Game iframe, About, Controls, session leaderboard, score form
Leaderboard	leaderboard.html	Tab per game, top-10 table, Clear Scores with confirmation
Submit a Game	submit.html	Validated form, 150-char counter, inline thank-you on submit

SECTION 04

Visual Design Specification

Token	Hex	Usage	Display face	Orbitron (Google Fonts, free)
Deep Space	#0D0F1A	Page background	Usage	H1, H2, logo, game titles only
Card Dark	#1A1D2E	Cards & panels	Body face	Inter (Google Fonts, free)
Neon Teal	#00E5CC	Accent, buttons, active	Usage	All paragraphs, labels, buttons
Soft White	#E8EAF0	Body text	Min body size	15px · line-height 1.6
Muted Grey	#6B7280	Secondary text	Breakpoints	640px mobile · 1024px tablet
Purple Badge	#7C3AED	Puzzle category	Max content W	1200px, centred via margin: 0 auto
Orange Badge	#EA580C	Arcade category	Card hover	translateY(-4px) + teal box-shadow
Blue Badge	#2563EB	Trivia category		

SECTION 05

Technical Specification

Folder Structure

Path	Contents / Notes
gamenest-hub/index.html	Homepage
gamenest-hub/catalog.html	Game catalog
gamenest-hub/game-detail.html	Game detail template (URL param: ?id={game_id})
gamenest-hub/leaderboard.html	Session leaderboard
gamenest-hub/submit.html	Game submission form
css/style.css	Single unified stylesheet no framework allowed
js/catalog.js	Filter, search, dynamic card rendering
js/game-detail.js	URL param read, iframe src, localStorage CRUD
js/leaderboard.js	Tab interface, score read, clear all scores
js/submit.js	Validation, char counter, thank-you state
games/{game-id}/	One local subfolder per game (9 total)
assets/thumbnails/	9 PNG thumbnails 400x225px, 16:9 ratio

localStorage Schema

Key format: **gamenest_scores_{game_id}**. Value: JSON array of {name: string, score: integer, date: ISO string}. On write: sort descending, keep top 10, JSON.stringify back. Clear: iterate Object.keys(localStorage) and delete all keys matching the prefix after browser window.confirm().

SECTION 06

Game Dataset (Public Domain / Open Source)

#	Title	Category	Licence	Repository / Source
1	2048	Puzzle	MIT	github: gabrielecirulli/2048
2	Hextris	Puzzle	GPL-3	github: Hextris/hextris
3	Color Flood	Puzzle	MIT	github: nicholasgasior/color-flood
4	Floppy Bird	Arcade	MIT	github: nebez/floppybird
5	Snake Game	Arcade	MIT	github: patorjk/snake
6	Breakout	Arcade	MIT	github: strd01/breakout
7	Trivia Quiz	Trivia	MIT	github: joshbучea/trivia
8	Open Trivia	Trivia	CC0	opentdb free public API, no key needed
9	Trivia Blitz	Trivia	MIT	github: guillaumebriday/trivia-app

SECTION 07

JavaScript Logic Summary

catalog.js	Games defined as a JS array at file top (id, title, category, description, thumbnail, detailPage). Cards rendered via innerHTML on load. Filter + search are event-driven, work together, update DOM with no page reload.
game-detail.js	Reads ?id= from URLSearchParams, matches against games array, sets iframe src to local games/{id}/ path. Leaderboard: read, parse, sort desc, render top 5. Score submit: validate name not empty + score is positive integer, push, sort, cap at 10, save, re-render.
leaderboard.js	Builds tabs dynamically from games array. Tab click reads that game's localStorage key and renders top 10 in ranked table (rank, name, score, date). Clear button deletes all gamenest_scores_* keys after window.confirm().
submit.js	Prevents default form submit. Validates all 5 fields shows inline red error below each empty field. Live 150-char counter on textarea shown as X / 150. On valid submit: hide form, show thank-you div no page reload, no backend call.

SECTION 08

Acceptance Criteria

#	Criterion	How to Verify
1	All 5 pages load without console errors.	Chrome DevTools Console tab zero errors.
2	Catalog filter + search work together.	Select Puzzle, type 'color' only matches show.
3	At least 6 of 9 games are playable in iframe.	Open 6 detail pages, interact with each game.
4	Scores persist and display on leaderboard.	Submit score, reload, confirm entry in table.
5	Submit form validates and shows thank-you inline.	Submit empty, then full observe both outcomes.
6	Responsive at 375px, 768px, 1280px.	DevTools device toolbar at each breakpoint.

#	Criterion	How to Verify
7	No CSS framework loaded.	Network tab no Bootstrap/Tailwind stylesheet.
8	No backend, no npm, no paid API in deliverable.	No package.json, no server file, no API keys.
9	Colours + fonts match design spec.	Compare rendered pages to reference PNG files.
10	All JS files use 2-space indentation. All functions and variables have descriptive names.	Open JS files, review readability manually.

SECTION 09

Deliverable Format

Submit all files as a single ZIP archive named **gamenest_hub_[your_name].zip** containing the full folder structure from Section 05. The project must open by launching index.html directly in Chrome, Firefox, Edge, or Safari with no build steps, no local server, and no external runtime dependencies. Thumbnails must be PNG at 400x225px. All 9 game subfolders must be present even if fewer than 9 games are playable.